In [ ]:
```
import os
import numpy as np
import seaborn as sns
import pandas as pd
import matplotlib.pyplot as plt
```

In [ ]:
```
path = os.path.abspath(os.getcwd())
file_path = os.path.join(os.path.dirname(path), "data", "rent_prediction.csv")

df = pd.read_csv(file_path)
```

Dimensions, colonnes et types

In [55]:
```
numeric_cols = list(df.drop("IdentifiantMaison", axis=1).select_dtypes(include=["number"]).columns)
category_cols = list(df.drop("IdentifiantMaison", axis=1).select_dtypes(include=["object", "str"]).columns

print(f"Colonnes dans les données : {list(df.columns)}")
print()
print(f"La forme de données du DataSet : {df.shape}")
print()
print(f"Type de données du DataSet : {df.info()}")
```

```
Colonnes dans les données : ['IdentifiantMaison', 'Chambres', 'Salon', 'SalleDeBainInterieure', 'Parking',
'Meuble', 'Jardin', 'Superficie_m2', 'DistanceRoute_m', 'Quartier', 'AgeMaison', 'LoyerMensuel_BIF']

La forme de données du DataSet : (510, 12)

<class 'pandas.DataFrame'>
RangeIndex: 510 entries, 0 to 509
Data columns (total 12 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   IdentifiantMaison      510 non-null    int64
 1   Chambres               485 non-null    float64
 2   Salon                  485 non-null    str
 3   SalleDeBainInterieure  485 non-null    str
 4   Parking                485 non-null    str
 5   Meuble                 485 non-null    str
 6   Jardin                 485 non-null    str
 7   Superficie_m2          485 non-null    float64
 8   DistanceRoute_m        485 non-null    float64
 9   Quartier               485 non-null    str
 10  AgeMaison              485 non-null    float64
 11  LoyerMensuel_BIF       510 non-null    int64
dtypes: float64(4), int64(2), str(6)
memory usage: 58.8 KB
Type de données du DataSet : None
```

Premier coup d'oeil aux données

In [13]:
```
df.head()
df.tail()
df.sample(10)
```

Out[13]:

	house_id	rooms	living_room	indoor_bathroom	parking	Meuble	Jardin	Superficie_m2	dstance_to_road	Quartier	h
478	479	3.0	Oui	Oui	Non	Non	Oui	164.0	300.0	Kiriri	
121	122	1.0	Oui	Non	Oui	Non	Non	36.0	49.0	Nyakabiga	
247	248	5.0	Oui	Oui	Non	Non	Oui	241.0	205.0	Ngagara	
361	362	5.0	Oui	Oui	Oui	Non	Oui	NaN	33.0	Rohero	
228	229	4.0	Oui	Oui	Oui	Non	Oui	208.0	186.0	Gasekebuye	
289	290	4.0	Oui	Oui	Oui	Non	Non	187.0	298.0	Kinanira	
59	60	4.0	Oui	Oui	NaN	Non	Non	NaN	209.0	Kiriri	
287	288	NaN	Oui	Oui	Non	Oui	Non	164.0	228.0	Gasekebuye	
310	311	3.0	Oui	Non	Oui	Non	Oui	119.0	26.0	Cibitoke	
259	260	1.0	Non	Oui	Oui	NaN	Non	55.0	224.0	Bwiza	

Statistiques descriptives

In [14]:
```
df.describe()
# df.describe(include=["object"])
```

Out[14]:

	house_id	rooms	Superficie_m2	house_age	monthly_rent
count	510.000000	485.000000	485.000000	4.850000e+02	5.090000e+02
mean	255.500000	3.521649	168.762887	5.380724e+03	1.241945e+06
std	147.368586	1.707436	79.610256	1.180590e+05	6.985334e+05
min	1.000000	1.000000	30.000000	0.000000e+00	1.361490e+05
25%	128.250000	2.000000	103.000000	1.000000e+01	6.991380e+05
50%	255.500000	4.000000	171.000000	2.000000e+01	1.077747e+06
75%	382.750000	5.000000	236.000000	3.000000e+01	1.678140e+06
max	510.000000	6.000000	320.000000	2.600000e+06	2.600000e+06

Identification de valeurs maquantes

In [15]:
```
df.isnull()
df.isnull().sum().sort_values(ascending=False)
```

Out[15]:
```
rooms              25
living_room        25
indoor_bathroom    25
Meuble             25
Jardin             25
Superficie_m2      25
dstance_to_road    25
Quartier           25
house_age          25
parking            24
monthly_rent        1
house_id            0
dtype: int64
```

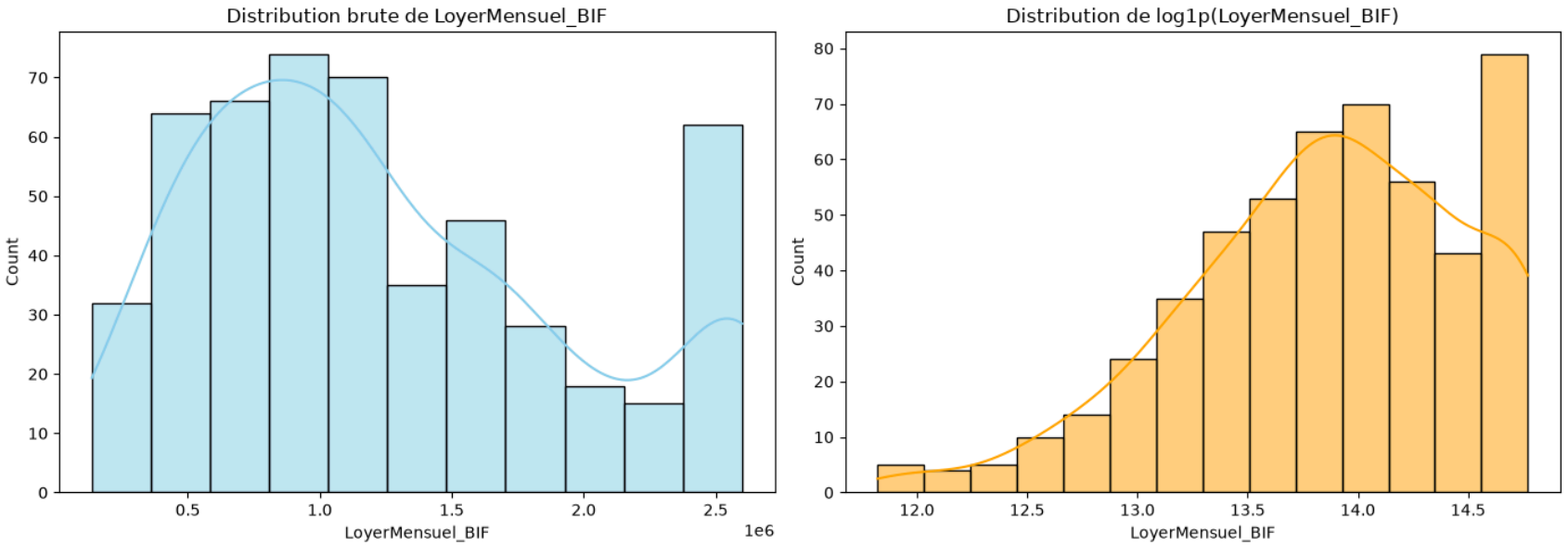
Lignes en double

In [17]:
```
df.duplicated()
df.duplicated().sum()
```

Out[17]:
```
0      False
1      False
2      False
3      False
4      False
       ...
505    False
506    False
507    False
508    False
509    False
Length: 510, dtype: bool
```

Analyse de la cible

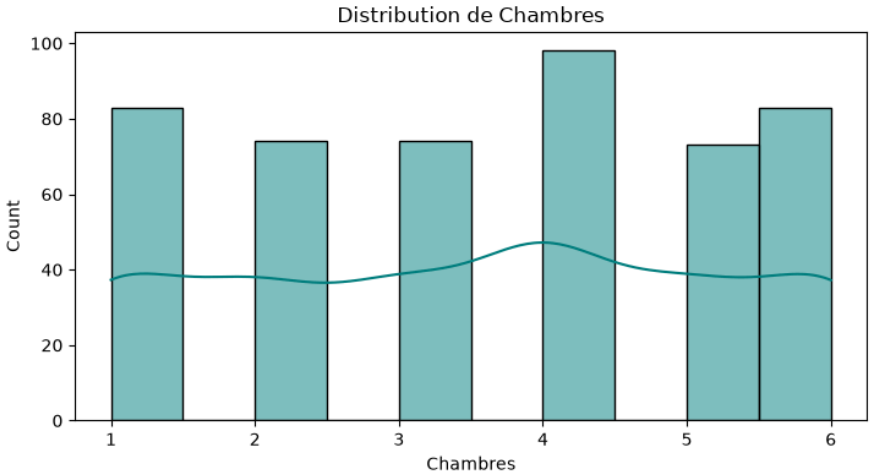
In [58]:
```
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

sns.histplot(df["LoyerMensuel_BIF"], kde=True, ax=axes[0], color="skyblue")
axes[0].set_title("Distribution brute de LoyerMensuel_BIF")

# Distribution avec Transformation Log
sns.histplot(np.log1p(df["LoyerMensuel_BIF"]), kde=True, ax=axes[1], color="orange")
axes[1].set_title("Distribution de log1p(LoyerMensuel_BIF)")

plt.tight_layout()
plt.show()
```

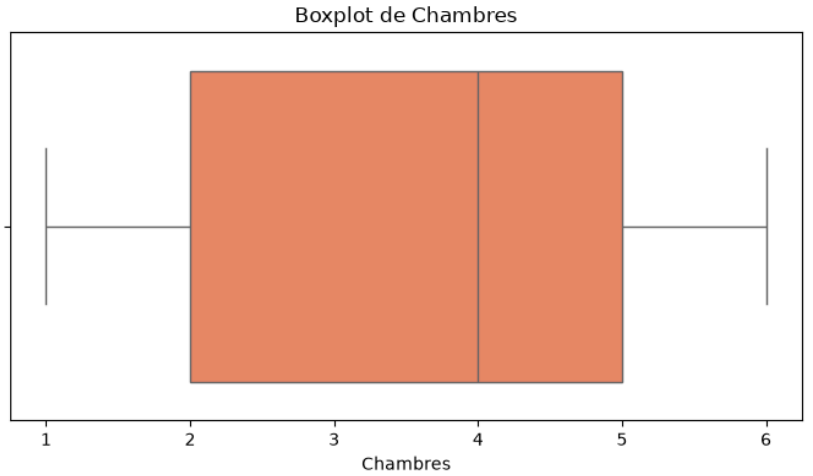


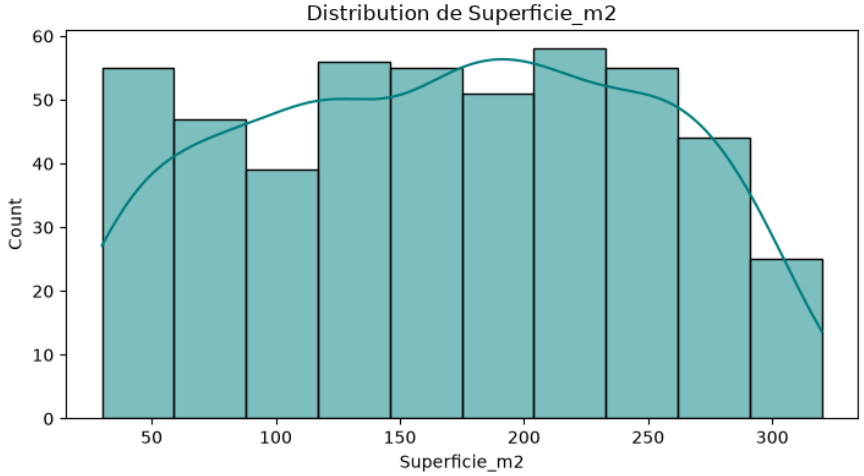
Analyses de features numériques

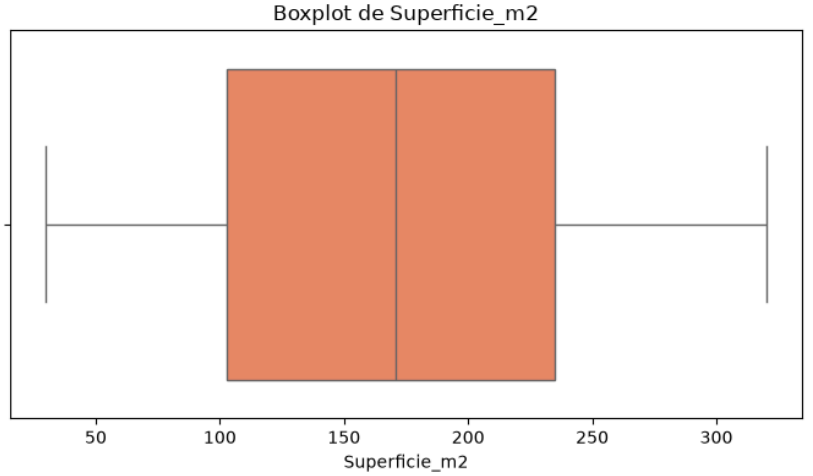
In [29]:
```
fig, axes = plt.subplots(len(numeric_cols), 2, figsize=(14, 4 * len(numeric_cols)))

for i, col in enumerate(numeric_cols):
    sns.histplot(df[col], kde=True, ax=axes[i, 0], color="teal")
    axes[i, 0].set_title(f"Distribution de {col}")

    sns.boxplot(x=df[col], ax=axes[i, 1], color="coral")
    axes[i, 1].set_title(f"Boxplot de {col}")
```

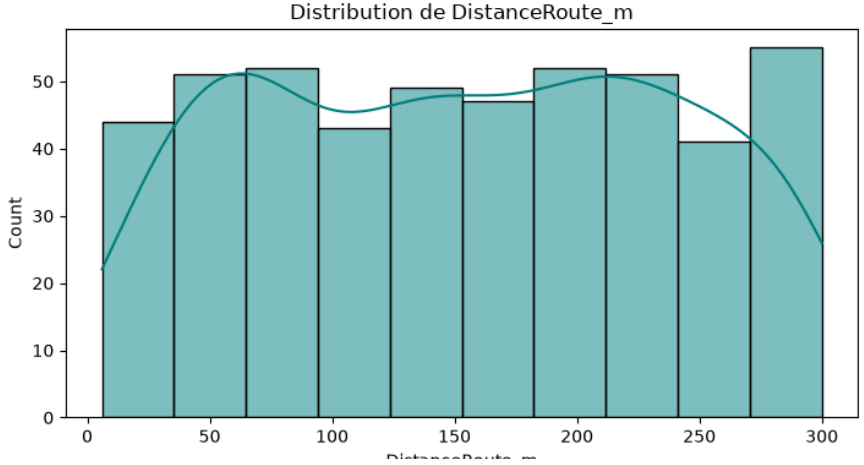
```
plt.tight_layout()
plt.show()
```

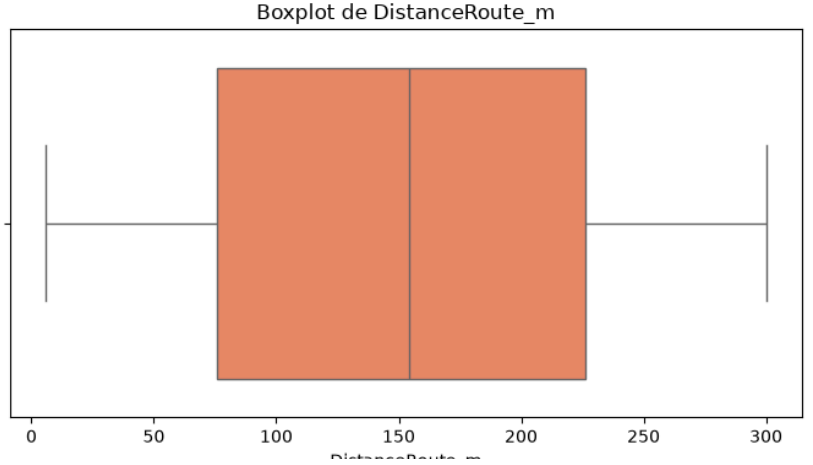

Distribution de Chambres
Count
Chambres

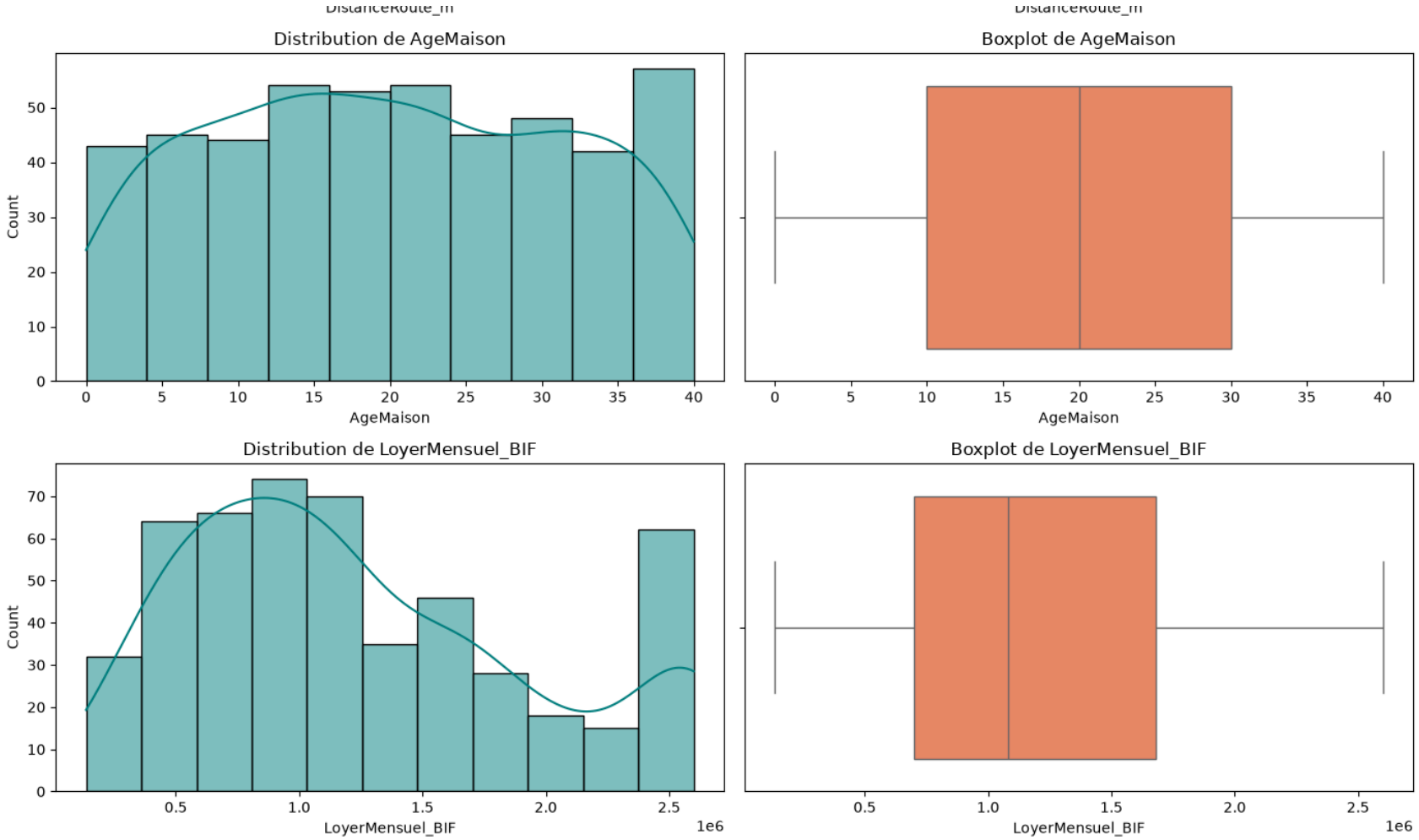

Boxplot de Chambres
Chambres


Distribution de Superficie_m2
Count
Superficie_m2


Boxplot de Superficie_m2
Superficie_m2


Distribution de DistanceRoute_m
Count


Boxplot de DistanceRoute_m



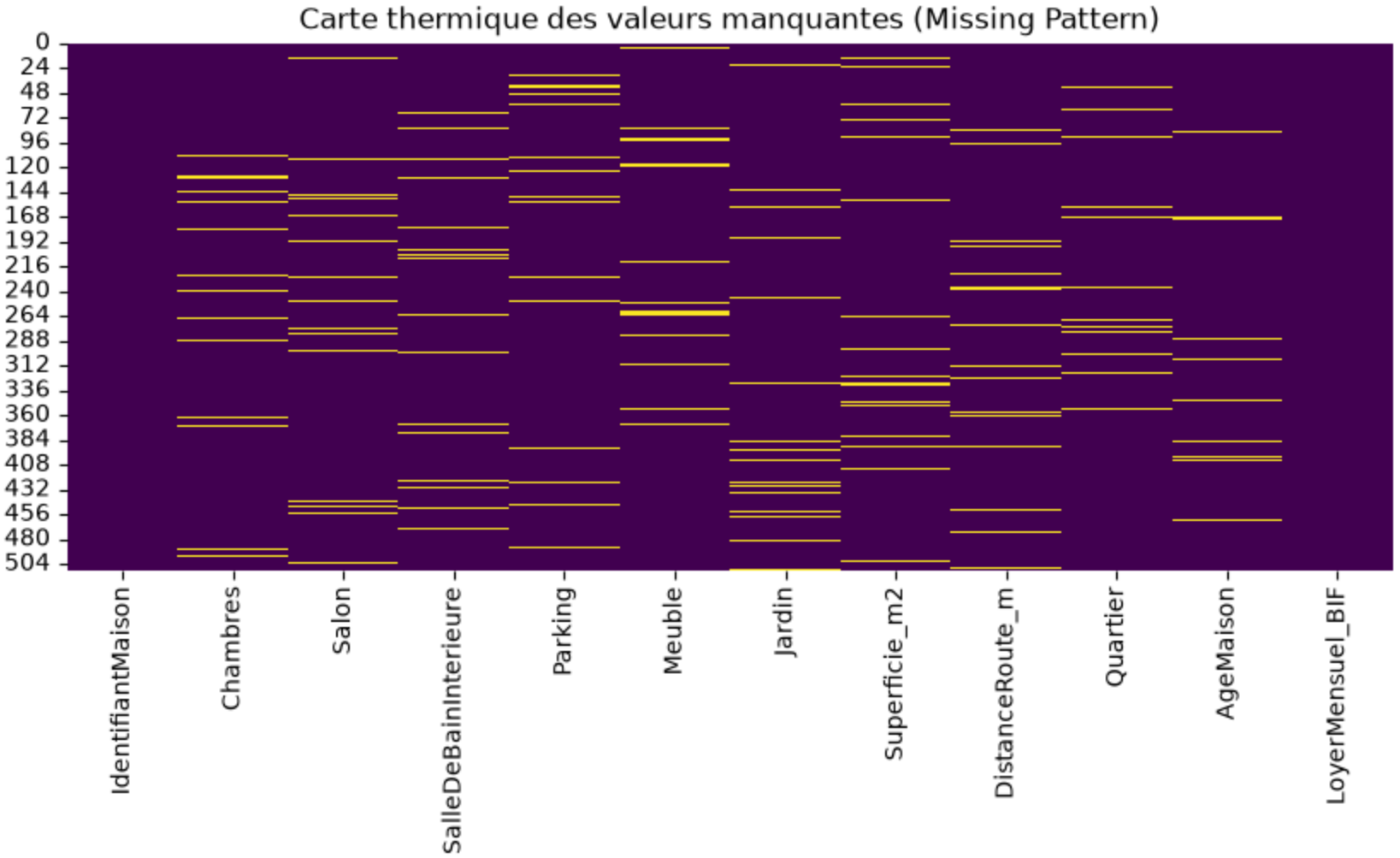
Analyses de variables catégorielles

In [ ]:
```
fig, axes = plt.subplots(3, 2, figsize=(14, 12))
axes = axes.flatten()

for i, col in enumerate(category_cols):
    sns.countplot(
        data=df,
        x=col,
        hue=col,
```

```
        legend=False,
        ax=axes[i],
        palette="Set2",
        order=df[col].value_counts().index,
    )
    axes[i].set_title(f"Répartition de {col} (Cardinalité: {df[col].nunique()})")
    axes[i].tick_params(axis="x", rotation=45 if df[col].nunique() > 5 else 0)

plt.tight_layout()
plt.show()
```

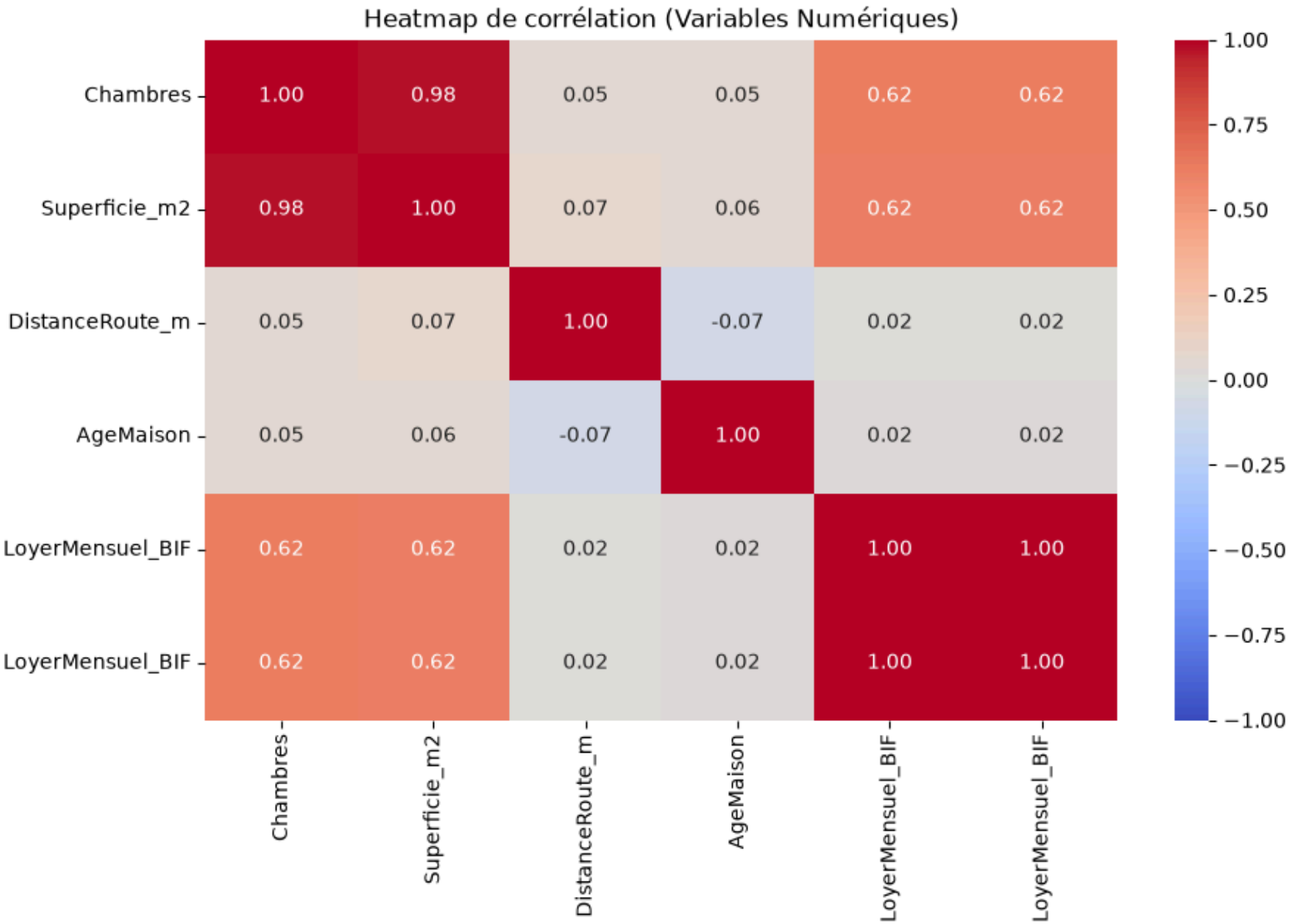
Pattern des valeurs maquantes

In [56]:
```
# Visualiser si les manques sont corrélés entre eux (MCAR vs MNAR)
plt.figure(figsize=(10, 4))
sns.heatmap(df.isnull(), cbar=False, cmap="viridis")
plt.title("Carte thermique des valeurs manquantes (Missing Pattern)")
plt.show()
```



Corrélation avec la variable cible (LoyerMensuel_BIF)

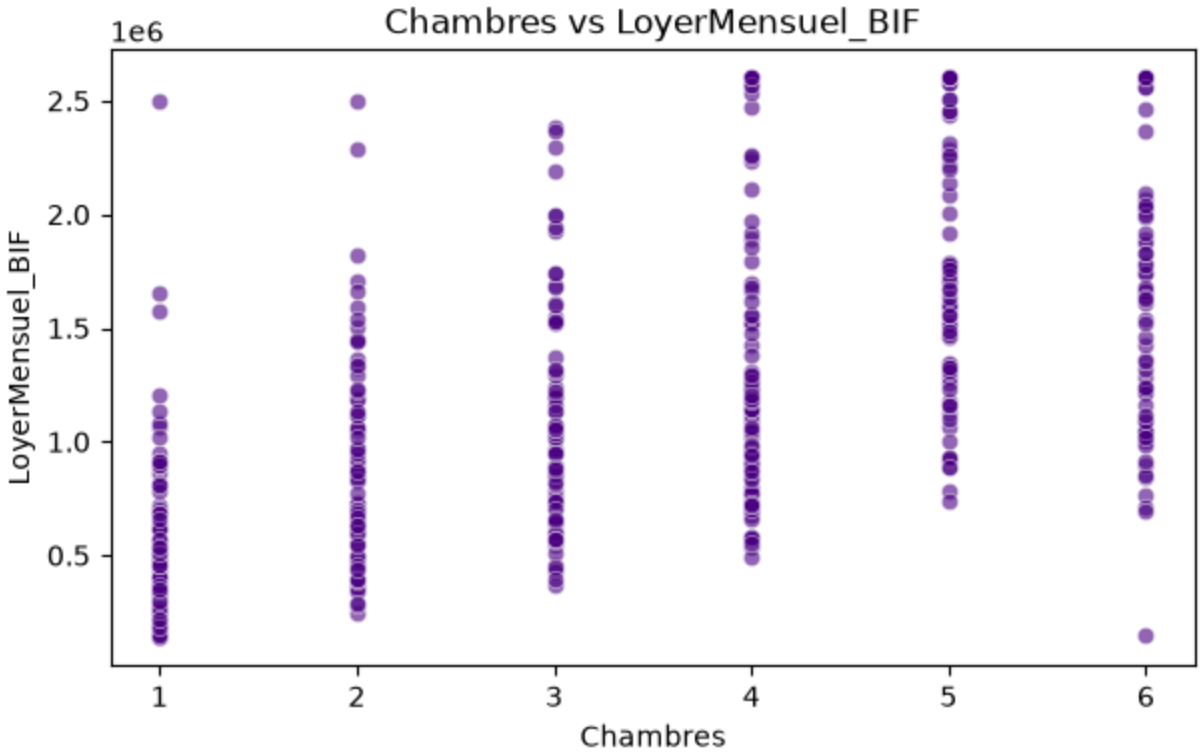
In [35]:
```
plt.figure(figsize=(1, 6))

corr_matrix = df[numeric_cols + ["LoyerMensuel_BIF"]].corr()
sns.heatmap(corr_matrix, annot=True, cmap="coolwarm", fmt=".2f", vmin=-1, vmax=1)
plt.title("Heatmap de corrélation (Variables Numériques)")
plt.show()
```

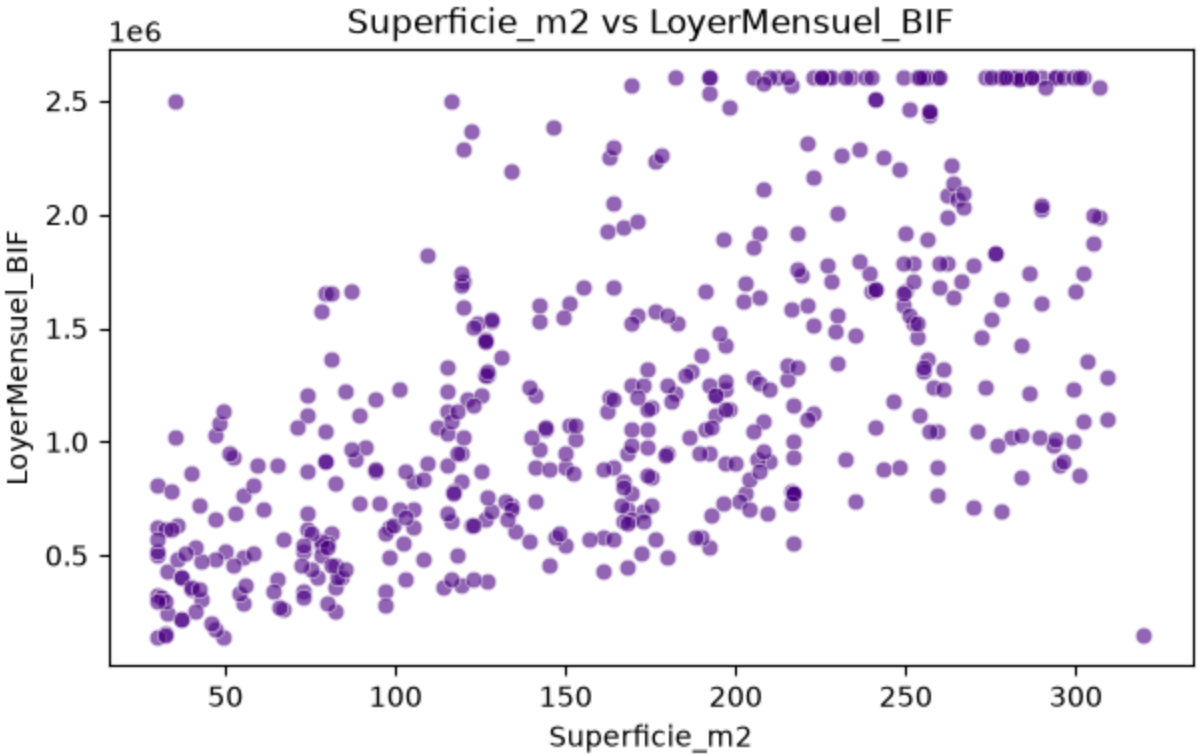


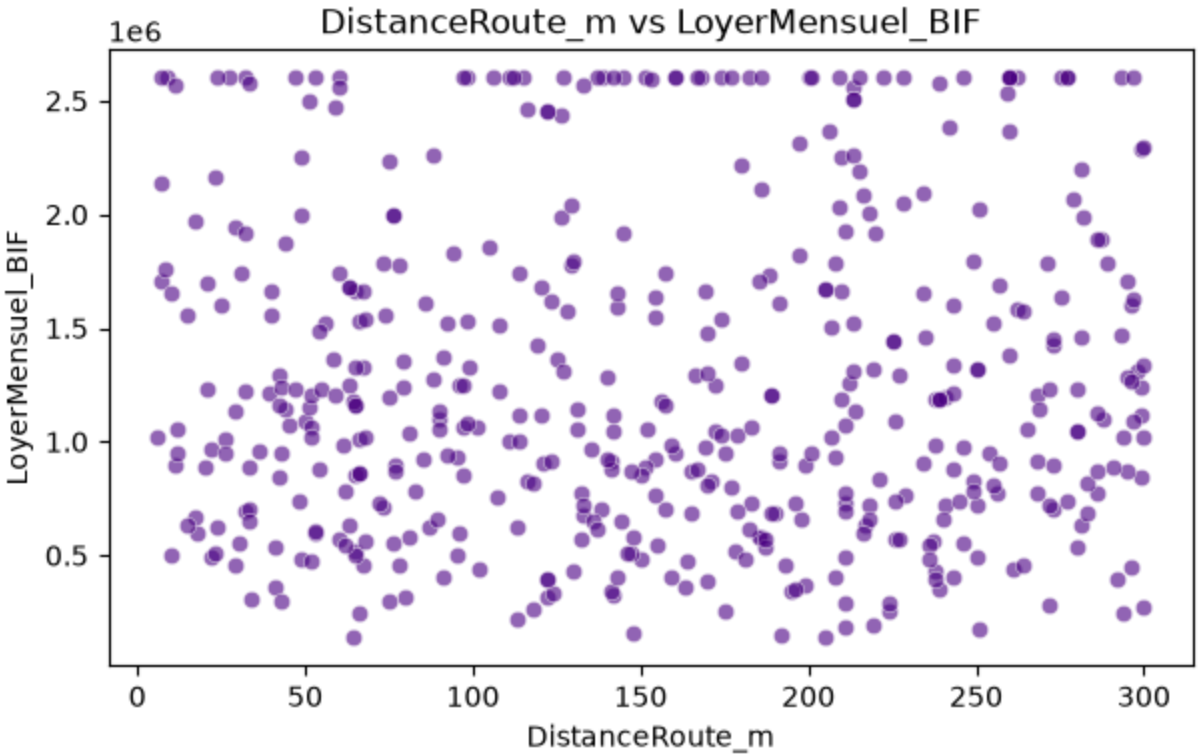
Scatter plots / Boxplots croisant chaque feature avec LoyerMensuel_BIF.

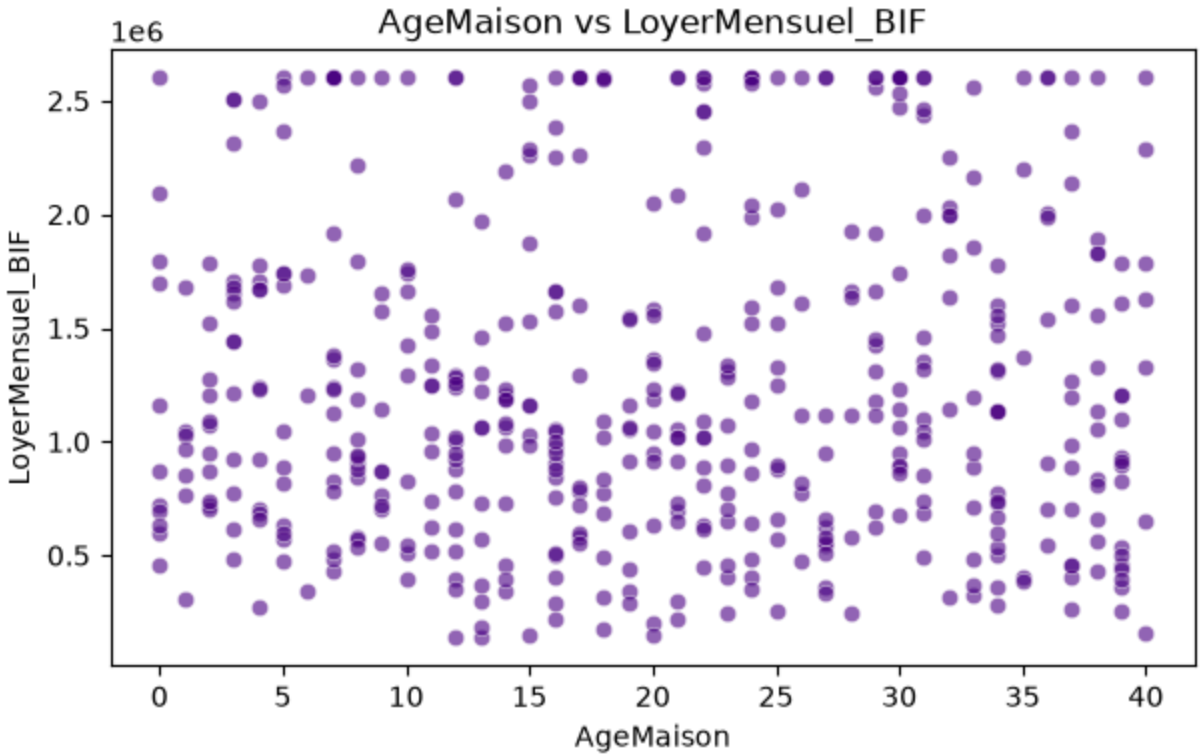
In [57]:
```
# Scatter plots pour les variables numériques
for col in numeric_cols:
```

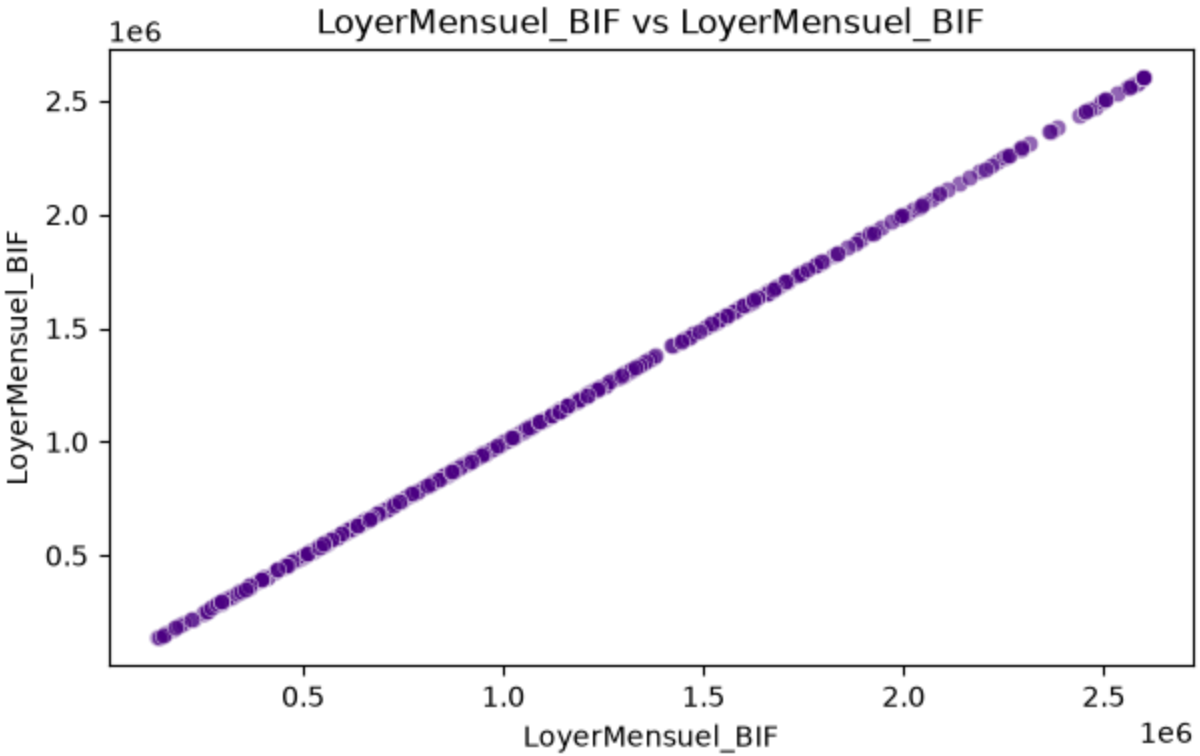
```
    plt.figure(figsize=(7, 4))
    sns.scatterplot(
        data=df, x=col, y="LoyerMensuel_BIF", alpha=0.6, color="indigo"
    )
    plt.title(f"{col} vs LoyerMensuel_BIF")
    plt.show()

# Boxplots pour les variables catégorielles
for col in category_cols:
    plt.figure(figsize=(8, 4))

    sns.boxplot(
        data=df,
        x=col,
        hue=col,
        legend=False,
        y="LoyerMensuel_BIF",
        palette="Pastel1",
        order=df.groupby(col)["LoyerMensuel_BIF"]
        .median()
        .sort_values(ascending=False)
        .index,
    )

    plt.title(f"LoyerMensuel_BIF selon {col}")
    plt.xticks(rotation=45 if df[col].nunique() > 5 else 0)
    plt.show()
```

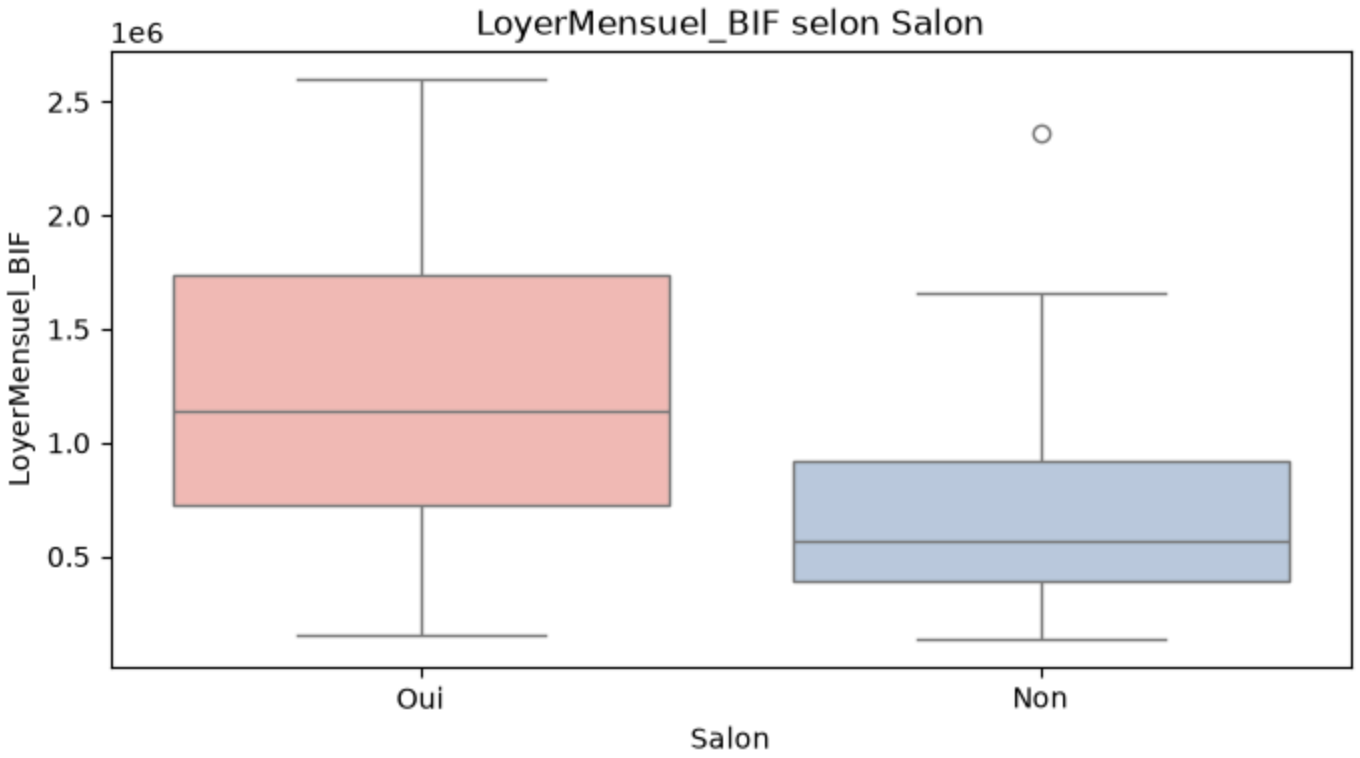

Chambres vs LoyerMensuel_BIF
1e6
2.5
2.0
1.5
1.0
0.5
LoyerMensuel_BIF
1
2
3
4
5
6
Chambres

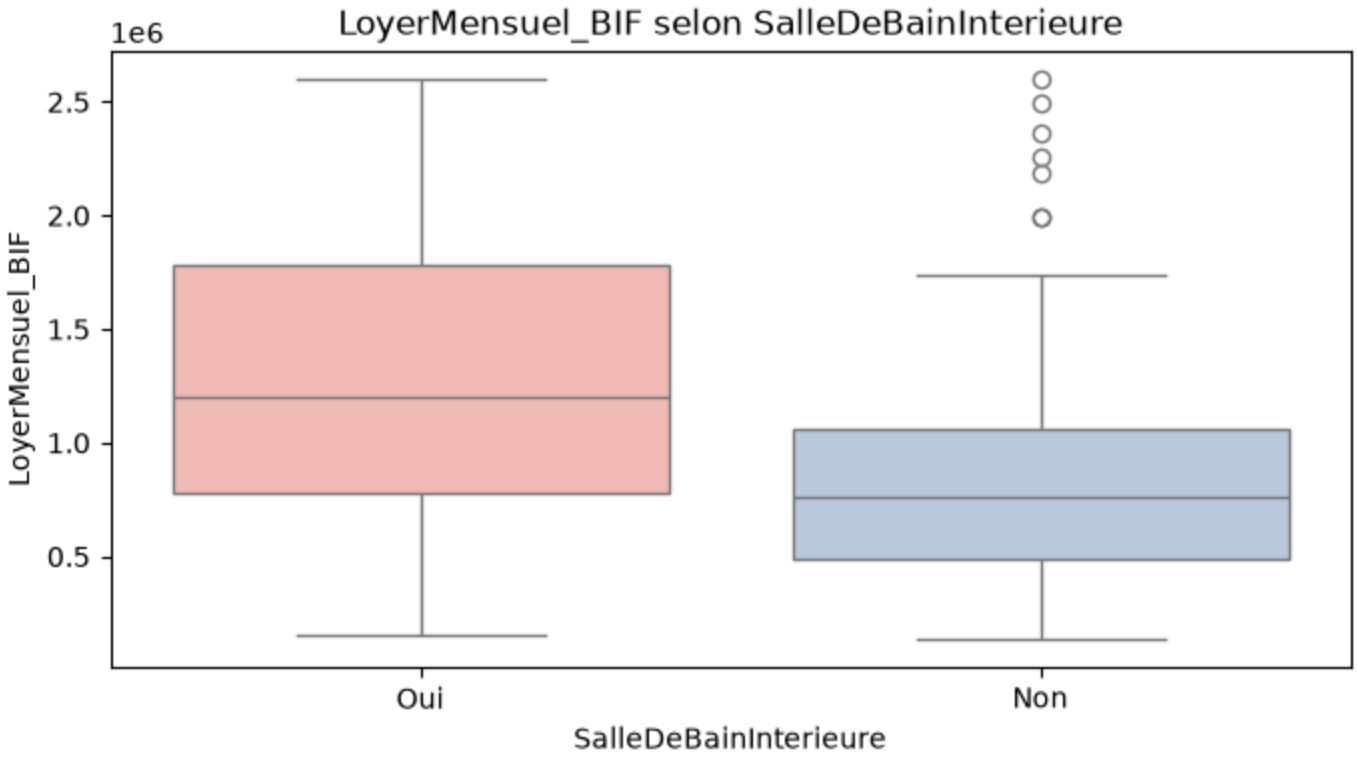

Superficie_m2 vs LoyerMensuel_BIF
1e6
2.5
2.0
1.5
1.0
0.5
LoyerMensuel_BIF
50
100
150
200
250
300
Superficie_m2

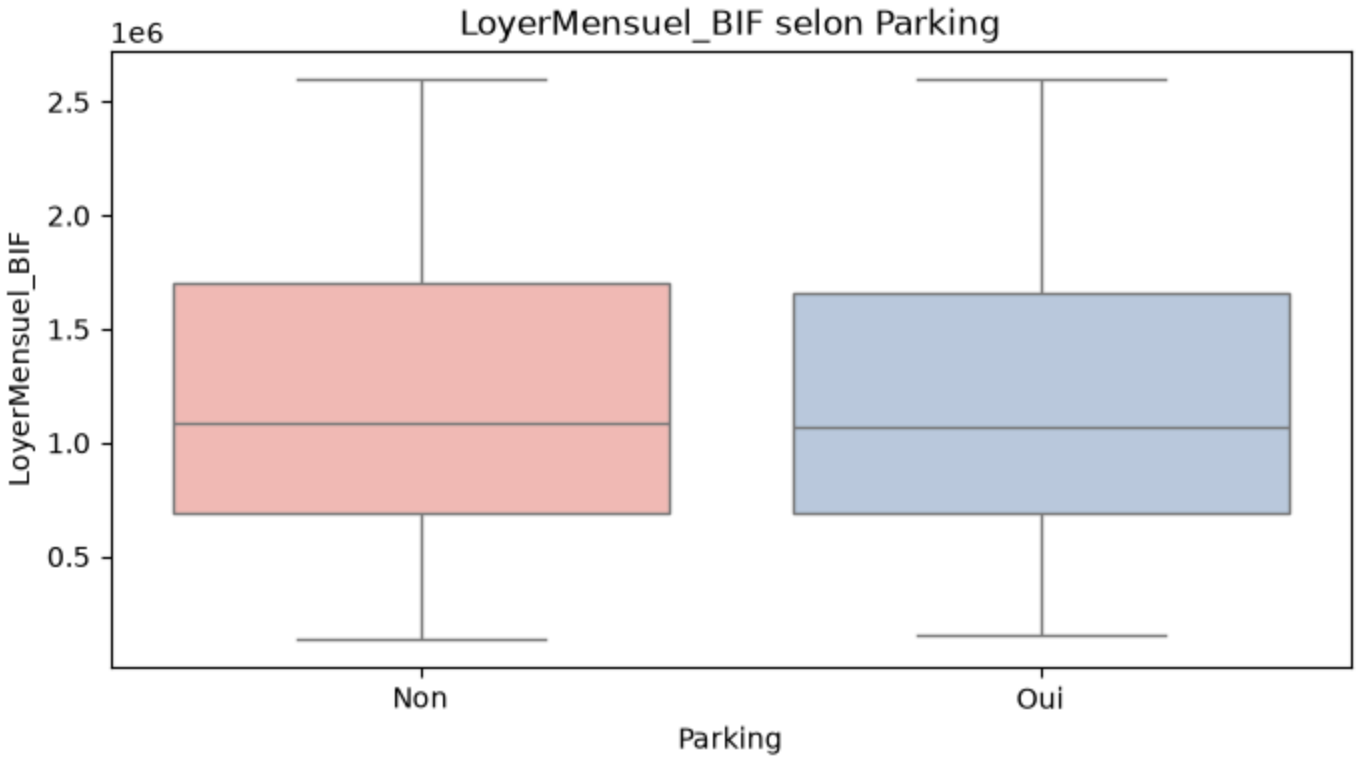

DistanceRoute_m vs LoyerMensuel_BIF
1e6
2.5
2.0
1.5
1.0
0.5
LoyerMensuel_BIF
0
50
100
150
200
250
300
DistanceRoute_m

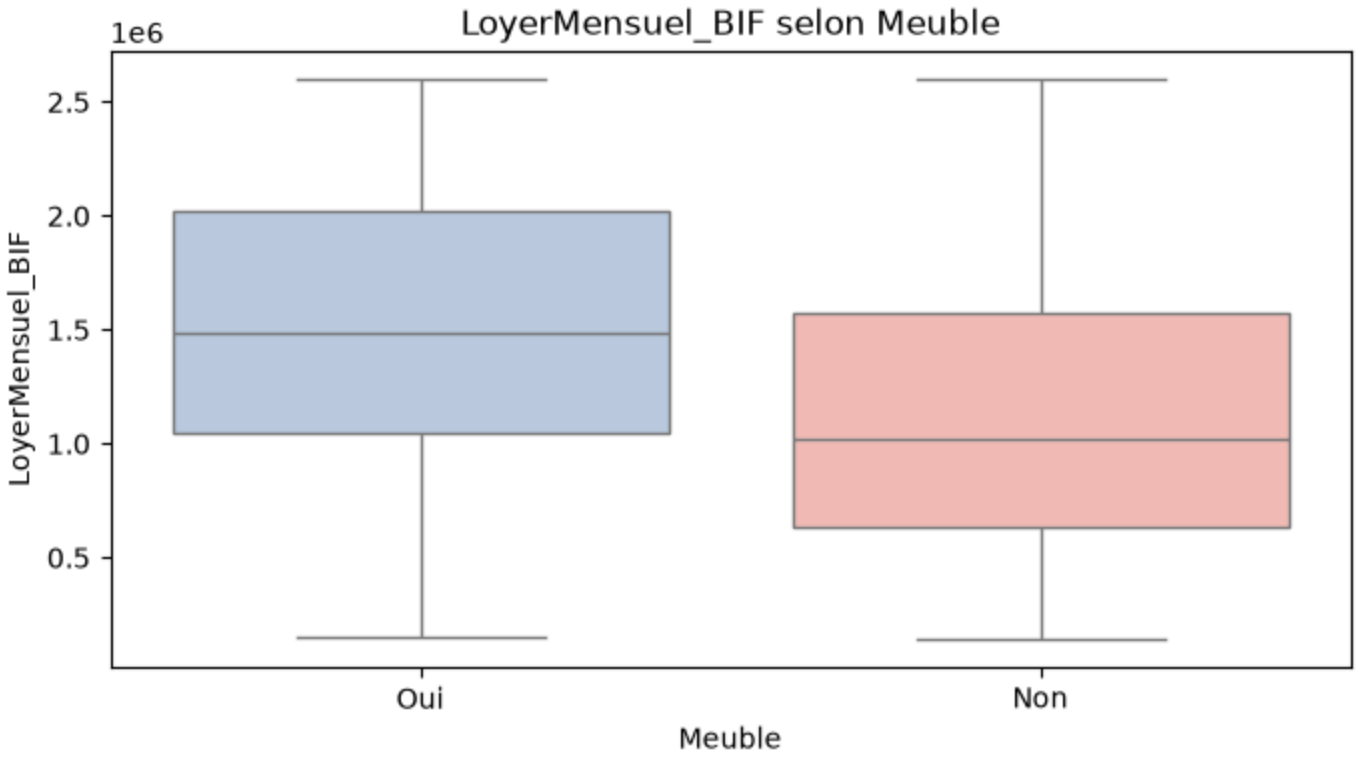

AgeMaison vs LoyerMensuel_BIF
1e6
2.5
2.0
1.5
1.0
0.5
LoyerMensuel_BIF
0
5
10
15
20
25
30
35
40
AgeMaison

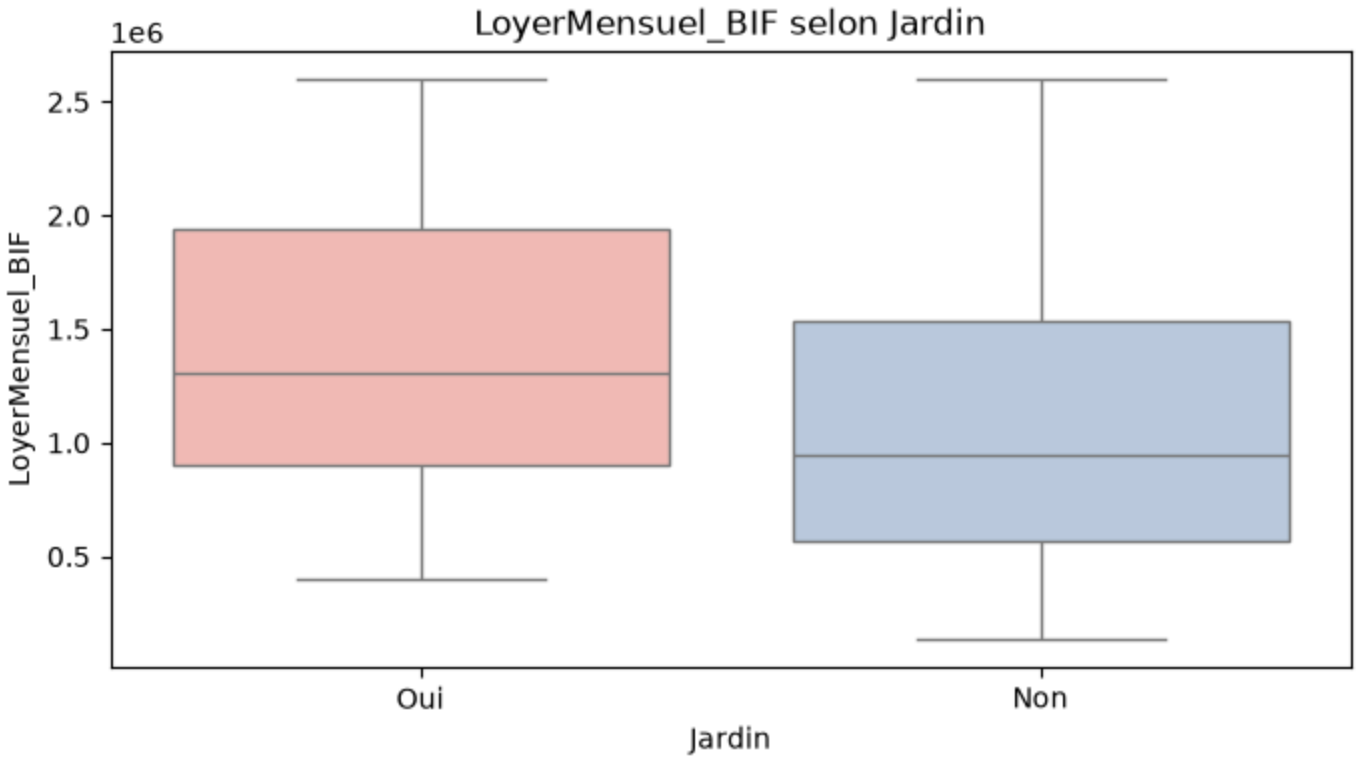

LoyerMensuel_BIF vs LoyerMensuel_BIF
1e6
2.5
2.0
1.5
1.0
0.5
LoyerMensuel_BIF
0.5
1.0
1.5
2.0
2.5
LoyerMensuel_BIF
1e6

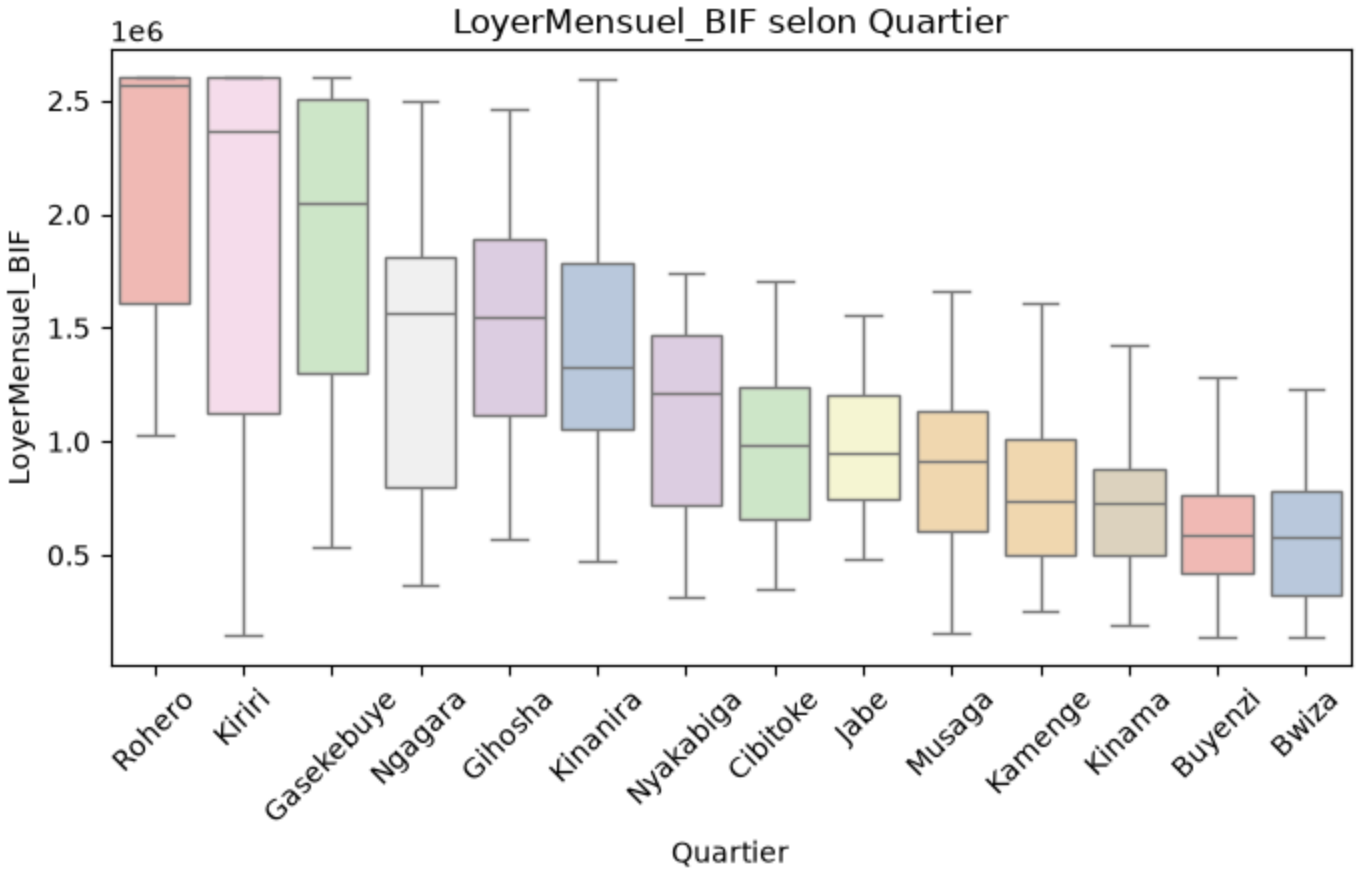

LoyerMensuel_BIF selon Salon
1e6
2.5
2.0
1.5
1.0
0.5
LoyerMensuel_BIF
Oui
Non
Salon


LoyerMensuel_BIF selon SalleDeBainInterieure
1e6
2.5
2.0
1.5
1.0
0.5
LoyerMensuel_BIF
Oui
Non
SalleDeBainInterieure


LoyerMensuel_BIF selon Parking
1e6
2.5
2.0
1.5
1.0
0.5
LoyerMensuel_BIF
Non
Oui
Parking


LoyerMensuel_BIF selon Meuble
1e6
2.5
2.0
1.5
1.0
0.5
LoyerMensuel_BIF
Oui
Non
Meuble


LoyerMensuel_BIF selon Jardin
1e6
2.5
2.0
1.5
1.0
0.5
LoyerMensuel_BIF
Oui
Non
Jardin


LoyerMensuel_BIF selon Quartier
1e6
2.5
2.0
1.5
1.0
0.5
LoyerMensuel_BIF
Rohero
Kiriri
Gasekebuye
Ngagara
Gihosha
Kinanira
Nyakabiga
Cibitoke
Jabe
Musaga
Kamenge
Kinama
Buyenzi
Bwiza
Quartier

In [ ]: